

RAHUL RAVI VK

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SUMMARY

Instrumentation & Controls Engineer with **6+ years** in safety-critical process industry, now widening expertise into Robotics, Computer Vision, and Industrial AI/ML through an M.Eng. in Robotics at the University of Maryland.

WORK EXPERIENCE

Research Intern | *Telehealth and Robotics Lab, Children's National Medical Center, Washington, D.C.* May 2026 – Aug 2026

- Conduct medical robotics and smart systems research under Dr. Reza Monfaredi (Sheikh Zayed Institute for Pediatric Surgical Innovation).
- Develop a smart wheelchair for children with Cerebral Palsy, a UR5-based surgical tool-sorting robot, and a UR3 telerobotic system driven by a Geomagic Touch haptic device.

Assistant Manager, Instrumentation & Controls | *KRIBHCO Fertilizers Ltd, Gujarat, India* May 2023 – Jul 2025

- Spearheaded automation and predictive maintenance across **Honeywell Experion PKS C300 DCS** and **Siemens PLC** arrays; engineered pump torque-converter speed control and PLC bistable-relay emulation, slashing equipment downtime **50%** and false trips **90%**.
- Orchestrated cross-functional cutovers (Mechanical, Electrical, Production, Civil) and directed annual shutdown/turnaround from scheduling through production restart.

Senior Engineer, Instrumentation & Controls | *KRIBHCO Fertilizers Ltd, Gujarat, India* Oct 2019 – Apr 2023

- Commanded DCS and safety-system operations supporting **4000+ control loops**; sustained **Honeywell Safety Manager (ESD)** and Experion DCS at **99.5%+** availability through RAID5 storage administration and one-to-one spare-station redundancy.
- Executed a 5-hour online tuning campaign on a safety-critical flow loop, eliminating oscillation and overshoot with zero production interruption; cut process downtime **30%** via PID retuning and actuator nonlinearity analysis.

Graduate Engineer Trainee, Instrumentation & Controls | *KRIBHCO Fertilizers Ltd, Gujarat, India* Oct 2018 – Oct 2019

- Established field-instrumentation foundation across live process lines: installed, calibrated, and validated 20+ transmitter types to $\pm 0.075\%$ FS across 4–20 mA / HART loops.
- Drove the full automation lifecycle: field wiring, calibration, PLC sequencing, safety interlocks, alarm rationalization, loop checking, FAT/SAT, and predictive maintenance.

PROJECTS

Autonomous Block Retrieval Robot *OpenCV, Raspberry Pi, Arduino, PID*

- Architected a fully autonomous robot from bare hardware to navigate a 10×10 ft arena, detect blocks in Red→Green→Blue sequence, and deliver **9/9 blocks in 7:21**; engineered HSV/contour detection, PID visual servoing, servo-gripper pickup confirmation, IMU–encoder fusion, and waypoint-based obstacle avoidance.

TurtleBot4 Real-Time Perception Pipeline *ROS 2, OpenCV, TurtleBot4, Classical CV*

- Engineered a ROS 2 vision-control pipeline delivering four real-time perception tasks: arrow path following, UMD logo stop, dynamic obstacle detection, and horizon overlay.
- Benchmarked deep-learning against classical computer-vision approaches and selected classical CV after evaluating on-robot deployment latency—validating AI solutions against real-world constraints.

Air Dryer Control Panel Redesign & Commissioning *Siemens PLC, TIA Portal, Ladder Logic*

- Redesigned and built an in-house PLC control panel to replace worn bistable-relay and mechanical-timer components, eliminating mechanical failure points from the air dryer regeneration path while preserving proven operator-familiar logic.
- Executed a single-day cutover within the 8-hour maintenance window in coordination with Mechanical, Electrical, Production, and Civil teams—transitioning to modern PLC-based control with zero production loss.

A* Path Planning with Pure Pursuit (Sim-to-Real) *ROS 2, Gazebo, TurtleBot3/4, Python*

- Implemented grid-based A* planning with a pure-pursuit follower from scratch; validated in Gazebo on TurtleBot3, then transferred to a physical TurtleBot4 with retuned velocity, lookahead, and heading-gain parameters, demonstrating end-to-end sim-to-real deployment.

EDUCATION

M.Eng. in Robotics | University of Maryland, College Park Sep 2025 – May 2027

GPA: 3.9/4.0 | Coursework: Perception, Planning, Control of Robotic Systems, Autonomous Robots, Robot Modeling, Industrial AI, ML for Perception & Sensing

B.Eng. in Instrumentation & Control | L.D. College of Engineering, Gujarat, India Aug 2014 – May 2018

GPA: 3.3/4.0 | GATE All India Rank 96

TECHNICAL SKILLS

- Robotics & Autonomy:** ROS 2, MoveIt 2, Gazebo, Nav2, URDF/Xacro, sim-to-real transfer, Linux
- Perception & AI/ML:** Computer vision, deep learning, image classification, sensor fusion, OpenCV, PyTorch
- Planning & Controls:** Motion planning (A*, RRT*), PID, cascade control, LQR/LQG, state estimation, Kalman filtering
- Embedded & Hardware:** Raspberry Pi, Arduino, NVIDIA Jetson, UART/I2C
- Industrial Automation:** DCS (Honeywell Experion PKS C300), Siemens PLC, SCADA, Safety Systems (ESD), FAT/SAT
- Programming & Tools:** Python, C++, MATLAB/Simulink, Git, Docker, SolidWorks

CERTIFICATIONS

Siemens – Basics of PLC & SCADA | Honeywell – Experion PKS C300 Fundamentals